

# MUSIC SOURCE SEPARATION VIA STEM DISCOVERY

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## ABSTRACT

Music source separation (MSS) methods aim to extract stems from music mixtures, which is important, for example, in karaoke, music remixing, and pedagogical applications. While earlier research on MSS systems has been dominated by models targeting narrow sets of general stems, there have recently been attempts to support broader source definitions. One of these methods uses audio queries to provide direct and descriptive control over the desired separation targets based on the sound itself. However, query-based separation remains cumbersome due to the need to provide audio examples with features matching the sources contained within mixtures. This paper proposes Music Source Separation via Stem Discovery (MuS3D), a query-based source separation framework that iteratively discovers active sources from the mixture. On correctly detected sources, our model matches manually queried baselines and surpasses state-of-the-art text-based models. Subjective evaluation indicates encoding artifacts as the limiting factor in current generative separation. The findings suggest that audio-based query representations offer an effective and automatable interface for source separation.

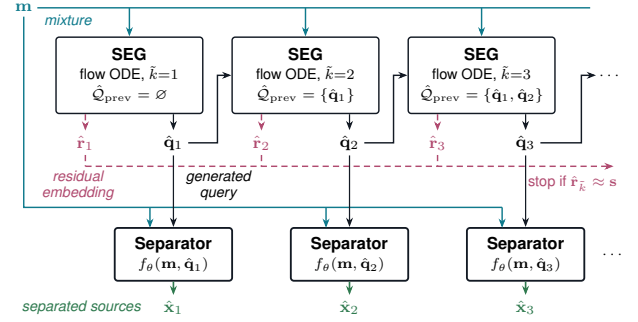
**Index Terms**— Audio systems, blind source separation, deep learning, music information retrieval

## 1. INTRODUCTION

Music source separation (MSS) refers to the task of dividing music mixtures into components that represent individual elements of the musical mix, such as vocals, percussion, or accompaniment instruments [1]. It is relevant to practical use cases such as karaoke track generation, music remixing and sampling, and pedagogy.

The dominant line of research on MSS has been dedicated to static settings with targets explicitly defined by the datasets as exemplified by recent research challenges [2, 3] with the goal of splitting the mix into four components: vocals, drums, bass, and other (VDBO). While a few systems have extended the static paradigm to additional instruments such as piano or guitar [4, 5], this rigid approach ultimately fails to scale. Many sound sources do not map cleanly onto a fixed instrument taxonomy, and naively growing the output vocabulary introduces severe target sparsity, since most instruments would be absent from any given mixture.

Recent studies have proposed models that offer the capability of separating arbitrary stems not strictly defined by the training data post hoc. Text-prompt-based systems rely on an additional text query that describes the sounds to be separated from the mixture [6, 7]. Similarly, audio queries provide a more semantically specific method of describing the desired target sounds [8, 9, 10]. Hyperbolic and hyperellipsoid queries instead provide region queries in the embedding space [11, 12], while multi-query systems combine modalities such as image, text, or sound [13, 14]. However, these methods require suitable manual queries, which are difficult to provide even with knowledge about the ground truth sources, limiting their use to, e.g., large-scale automated MSS.



**Fig. 1.** Block diagram of the proposed MuS3D framework for the first three iterations, each of which separates one source or stem group.

We denote systems that isolate all constituent sources in a mixture without prior knowledge of their identities or count as *blind*. Alternative blind approaches either use a predetermined number of outputs [15], inherit the limitations of static output vocabularies [16], or use a combination of large language models (LLMs) to describe the sources in the mixture, requiring accurate textual descriptions [17]. While these methods were developed for speech or general sound scenes, no prior work addresses blind separation for music, where sources exhibit strong temporal and harmonic correlations.

Motivated by these shortcomings, this paper proposes Music Source Separation via Stem Discovery (MuS3D)<sup>1</sup>. As illustrated in Fig. 1, MuS3D uses a stem embedding generator (SEG), also introduced in this work, to generate a set of embeddings for the constituent stems in the music mixture. These embeddings then serve as conditioning queries for a source separator, enabling it to extract each target sound from the mix. Experiments demonstrate that MuS3D separates music mixtures into fine-grained instrument stems without requiring external side information or manual queries.

The rest of the paper is organized as follows. Section 2 describes the MuS3D framework. Section 3 details the experimental setup. Section 4 outlines the results from objective metric and subjective listening evaluations. Section 5 concludes the paper. Sound examples for separated stems are available online.<sup>2</sup>

## 2. METHODS

Formally, source separation describes the task of decomposing a mixture  $\mathbf{m} = \sum_{k=1}^K \mathbf{x}_k$  into its  $K$  underlying components  $\mathbf{x}_k \in \mathbb{R}^{C \times T}$ , where  $C$  denotes the number of channels and  $T$  the number of samples. In MSS, the components  $\mathbf{x}_k$  generally consist of vocal or instrumental tracks, or stem groups thereof [1]. These tracks or stems could contain applied audio effects, such as reverberation or dynamic range compression, such that their summation exactly reconstructs

<sup>1</sup>Pronounced like *mused*.

<sup>2</sup><http://research.spa.aalto.fi/publications/papers/mus3d>

**Algorithm 1** Training of Stem Embedding Generator

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**Require:** feature extractor  $\psi$ , stop symbol  $\mathbf{s}$ , parameters  $\phi$

- 1: **repeat**
- 2:   Sample  $\{\mathbf{x}_1, \dots, \mathbf{x}_K\}$  from dataset in random order
- 3:    $\mathbf{m} \leftarrow \sum_{k=1}^K \mathbf{x}_k$  ▷ Obtain mixture
- 4:    $\tilde{k} \sim \mathcal{U}\{1, \dots, K\}$  ▷ Truncate the recursion
- 5:    $\mathbf{q} \leftarrow \psi(\mathbf{x}_{\tilde{k}})$ ,  $\mathcal{Q}_{\text{prev}} \leftarrow \{\psi(\mathbf{x}_k)\}_{k < \tilde{k}}$
- 6:    $\mathbf{r} \leftarrow \psi(\mathbf{m} - \sum_{k=1}^{\tilde{k}} \mathbf{x}_k)$  if  $\tilde{k} < K$  else  $\mathbf{s}$
- 7:    $\mathbf{z}_0 \leftarrow \text{concat}(\mathbf{q}, \mathbf{r})$ ,  $\mathbf{z}_1 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ ,  $t \sim \mathcal{U}[0, 1]$
- 8:    $\mathbf{z}_t \leftarrow t \mathbf{z}_1 + (1 - t) \mathbf{z}_0$  ▷ Linear interpolant
- 9:    $\mathcal{L} \leftarrow \|u_\phi(\mathbf{z}_t; t, \tilde{k}, \mathbf{m}, \mathcal{Q}_{\text{prev}}) - (\mathbf{z}_1 - \mathbf{z}_0)\|^2$  ▷ CFM loss
- 10:    $\phi \leftarrow \text{AdamW}(\phi, \nabla_\phi \mathcal{L})$
- 11: **until** converged

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the original mixture.

### 2.1. Query-based source separation

Query-based source separators take as input the time-domain signal  $\mathbf{m}$  and a query  $\mathbf{q}_k$  that describes the source  $\mathbf{x}_k$  to separate. The output is an estimate  $\hat{\mathbf{x}}_k$  of the target source signal:

$$\hat{\mathbf{x}}_k = f_\theta(\mathbf{m}, \mathbf{q}_k). \quad (1)$$

Conceptually, an ideal query maximizes mutual information with its target  $I(\mathbf{q}_k; \mathbf{x}_k)$  while minimizing it with interfering sources  $I(\mathbf{q}_k; \mathbf{x}_j)$  for all  $j \neq k$ . While we do not enforce this criterion explicitly, it serves as a design principle when choosing the query representation. We define the complete query set  $\mathcal{Q} = \{\mathbf{q}_k\}_{k=1}^K$  and implement the queries  $\mathbf{q}_k \in \mathbb{R}^Q$  as time-invariant embeddings obtained from a feature extractor  $\mathbf{q}_k = \psi(\mathbf{x}_k)$ , following prior conventions [9].

### 2.2. Recursive stem embedding generation

In the blind MSS scenario, both the number of sources  $K$  and the query set  $\mathcal{Q}$  are unknown *a priori*. We propose estimating the query set recursively with a flow model  $u_\phi$  producing one query embedding at a time, conditioned on the mixture and the previously generated queries. We refer to this as SEG.

Inference is illustrated in Fig. 1. At iteration  $\tilde{k}$ , the SEG model generates a pair  $\mathbf{z}_0 = [\mathbf{q}_{\tilde{k}}, \mathbf{r}_{\tilde{k}}]$ : the query for the next source  $\mathbf{q}_{\tilde{k}}$ , and a *residual embedding*  $\mathbf{r}_{\tilde{k}}$  describing the sources that remain. The pair is obtained by solving the flow Ordinary Differential Equation (ODE) [18] from  $t = 1$  to  $t = 0$ ,

$$d\mathbf{z}_t = u_\phi(\mathbf{z}_t, t, \tilde{k}, \mathbf{m}, \hat{\mathcal{Q}}_{\text{prev}}) dt, \quad (2)$$

where  $u_\phi$  is a Transformer [19] with parameters  $\phi$  trained by conditional flow matching (CFM) [18], and  $\hat{\mathcal{Q}}_{\text{prev}} = \{\hat{\mathbf{q}}_k\}_{k < \tilde{k}}$  is the set of queries generated so far.

Generating the residual alongside the query makes the recursion self-terminating. When no sources remain, the model is trained to emit a fixed *stop symbol*  $\mathbf{s}$  in the residual slot instead of an embedding of audio. The recursion halts once  $\langle \hat{\mathbf{r}}_{\tilde{k}} / \|\hat{\mathbf{r}}_{\tilde{k}}\|, \mathbf{s} \rangle > \tau_{\text{stop}}$ .

Training is described in Algorithm 1. Each example reconstructs one step of the recursion: we shuffle the stems of a track, draw a depth  $\tilde{k}$  uniformly, and ask the model to produce  $\psi(\mathbf{x}_{\tilde{k}})$  given the mixture and the descriptors of the preceding stems, with the residual target set to  $\mathbf{r}_{\tilde{k}} = \psi(\mathbf{m} - \sum_{k=1}^{\tilde{k}} \mathbf{x}_k)$ , or to  $\mathbf{r}_{\tilde{k}} = \mathbf{s}$  when  $\tilde{k} = K$ . Sampling the depth uniformly means a single pass supervises every stage of the recursion, including its termination, without ever unrolling it:

the cost of a training step is independent of  $K$ . Because the stem order is shuffled at training time, no iteration index is associated with any particular instrument, and the order in which queries emerge at inference is expected to be uniformly random.

## 3. EXPERIMENTS

We base our experiments on the MoisesDB dataset [20], using the splits from [9]. MoisesDB organizes sources into 11 broad and 38 fine categories, denoted *groups* and *instruments*. We train all models with both levels, choosing one at random and indicating it with a binary hierarchy-level flag  $h$ . The training set is supplemented with 125 songs from MedleyDB [21] and 10 from AlbumDB [22], organized to match the MoisesDB taxonomy. We use wet, processed stems and exclude MoisesDB songs with bleeding tracks. In total, 279 songs are used for training, 48 for testing, and 26 for validation.

For each training example, we randomly sample a song and extract a 10-s segment, loading its aligned stems from either the group or instrument hierarchy level. At the *group* level, stems from the same taxonomy group are summed; at the *instrument* level, we merge stems sharing a taxonomy leaf. We always sum all vocal tracks into one unit. We discard near-silent segments or those with fewer than two active stems based on RMS and relative loudness thresholds, and randomly drop some stems as augmentation. Each stem is also augmented with random gains and randomized effect chains (filtering, EQ, convolution reverb, compression, etc.); details are available online.<sup>3</sup>

To build the test set, each of the 48 test songs is split into non-overlapping 10-s segments on a fixed grid, giving 1062 segments in total. Group and instrument-level stems are defined as in training. No augmentations are applied. A source counts as active in a segment, and thus enters the target set  $\mathcal{S}$ , when its RMS is at least  $-50$  dBFS and no more than 35 dB below that of the mixture. Segments contain, on average, 4.5 active groups (1 to 8) or 7.0 instruments (1 to 15).

### 3.1. Query descriptor

As in [9, 12], the experiments employ the PaSST model [23] trained on OpenMIC-2018 [24]. The model uses 10-s long audio clips summed to mono and downsampled to 32 kHz as input to generate the embeddings. Additionally, following [12] we apply principal component analysis (PCA) using samples from the training datasets to reduce the feature dimension of the embeddings to 128 with a total explained variance of 0.96.

### 3.2. Stem embedding generator

The SEG architecture is a diffusion transformer (DiT) with 12 layers, a hidden size of 384, and 12 attention heads, totaling roughly 37M parameters. It is trained on the PCA-reduced 128-dimensional PaSST embeddings, producing one query at a time (see Fig. 1). The mixture  $\mathbf{m}$ , encoded both as its PCA-reduced PaSST embedding and as a sequence of DAC-VAE latents [14] (25 Hz, 128-dimensional), conditions the DiT via cross-attention, together with the previously generated queries  $\hat{\mathcal{Q}}_{\text{prev}}$ . The flow time  $t$ , the step counter  $\tilde{k}$  and the hierarchy-level flag  $h$  condition the model via adaptive layer normalization. All models are trained on a single NVIDIA H200 GPU. SEG is trained for 400k iterations with a batch size of 8.

At inference we solve (2) with a stochastic flow sampler [25], using  $T = 32$  uniform steps, and a constant churn parameter  $\gamma = 0.1$ .

<sup>3</sup>github.com/REMUS-Aalto/Mus3D (will be available upon acceptance)

The *stop symbol* of Section 2.2 is the least-variance direction of the PCA-reduced space, leaving a wide margin to any real source embedding, since this direction carries almost no energy for real sources. We use a threshold of  $\tau_{\text{stop}} = 0.9$ , since it did not cause premature stopping for the recursion. However, as a fallback mechanism, we cap the maximum number of iterations at 16.

### 3.3. Query-based separator models

*Diffusion transformer separator* (DiT-Sep): We implement a query-based separator also based on a CFM-trained DiT, building upon the *large* version of SAM-Audio [14], consisting of roughly 2.2B trainable parameters.<sup>4</sup> We adapt the original implementation by conditioning the transformer, via adaptive layer normalization, on the flow time  $t$ , the 128-dimensional query embeddings, and the hierarchy flag  $h$ . We initialize the training with the public weights<sup>5</sup>, and fine-tune it for roughly 300k iterations at a batch size of 16, using our train dataset, tracking the exponential moving average of the weights with a rate of 0.9999. At inference we use the stochastic flow sampler of [25] with  $T = 32$  uniform steps and a constant churn parameter  $\gamma = 0.5$ .

*BS-LoCoformer separator*: We use the musical band-split (BS) variant of the time-frequency (TF)-LoCoformer [26], in the *medium* setting, with query conditioning via bias modulation of the SwiGLU activations. Models train for approximately 200k iterations at a batch size of 4.

## 4. EVALUATION

We first assess how accurately SEG discovers the stems in a mixture (Section 4.1), and then evaluate separation quality with the generated queries against state-of-the-art MSS baselines using objective (Section 4.2) and subjective experiments (Section 4.3).

### 4.1. Stem discovery

Although SEG generates continuous, open-set embeddings, we evaluate stem discovery as a multi-label classification task over the MoisesDB taxonomy, both at *group* and *instrument* levels, allowing a direct comparison with baselines that predict instrument labels.

To assign a label to each generated query, we train a hierarchical stem classifier on the PCA-reduced PaSST embeddings, following [27]: a linear layer with sigmoid activations per instrument tag, with group activations obtained by max pooling over their tags, trained with a weighted binary cross-entropy loss at both levels on individual stems and group sums. Each query is assigned the highest-activation tag. Since the evaluation is bounded by the classifier accuracy, we include an Oracle classifier operating directly on ground truth (GT) stem embeddings to provide an upper bound for SEG.

We compare SEG against two baselines that predict the stems directly from the mixture. The first is a multi-label classifier with the same architecture and loss as the stem classifier, trained on the PCA-reduced PaSST embeddings of the SEG training mixtures to predict all their active tags. It outputs every tag whose activation exceeds a tag-specific threshold, chosen to maximize the  $F_1$  of that tag on the validation set. The second is MusicFlamingo [28], a language model for music understanding, which we prompt to list the stems

<sup>4</sup>The original SAM-Audio large implementation has 3B parameters, but we did not use the cross-attention mechanism.

<sup>5</sup>According to [14], MoisesDB was not used to train SAM-Audio, so the test set is not seen by the pretrained model.

**Table 1.** Objective evaluation of stem discovery, scored per 10-s segment and aggregated over the 1062 test segments (mean  $\pm$  std.). The best result in each column is highlighted in bold.

|                   | Method                    | $F_1$                  | Precision              | Recall                 | $\Delta K$             |
|-------------------|---------------------------|------------------------|------------------------|------------------------|------------------------|
| <i>All Groups</i> | Oracle Classifier         | 0.96 $\pm$ 0.10        | 0.96 $\pm$ 0.10        | 0.96 $\pm$ 0.10        | 0.00 $\pm$ 0.00        |
|                   | Multi-label Classifier    | <b>0.89</b> $\pm$ 0.12 | 0.88 $\pm$ 0.15        | <b>0.91</b> $\pm$ 0.14 | 0.69 $\pm$ 0.82        |
|                   | MusicFlamingo             | 0.82 $\pm$ 0.19        | 0.87 $\pm$ 0.21        | 0.80 $\pm$ 0.22        | 1.04 $\pm$ 1.07        |
|                   | SEG                       | 0.88 $\pm$ 0.13        | <b>0.92</b> $\pm$ 0.14 | 0.86 $\pm$ 0.16        | <b>0.67</b> $\pm$ 0.78 |
|                   | SEG (with res. embedding) | 0.85 $\pm$ 0.14        | 0.87 $\pm$ 0.16        | 0.85 $\pm$ 0.16        | 0.71 $\pm$ 0.81        |
| <i>All Instr.</i> | Oracle Classifier         | 0.86 $\pm$ 0.16        | 0.86 $\pm$ 0.16        | 0.86 $\pm$ 0.16        | 0.00 $\pm$ 0.00        |
|                   | Multi-label Classifier    | <b>0.75</b> $\pm$ 0.16 | 0.70 $\pm$ 0.19        | <b>0.84</b> $\pm$ 0.17 | 2.01 $\pm$ 1.80        |
|                   | MusicFlamingo             | 0.33 $\pm$ 0.14        | 0.44 $\pm$ 0.19        | 0.27 $\pm$ 0.14        | 3.17 $\pm$ 2.21        |
|                   | SEG                       | 0.74 $\pm$ 0.18        | <b>0.77</b> $\pm$ 0.20 | 0.72 $\pm$ 0.19        | <b>1.26</b> $\pm$ 1.18 |
|                   | SEG (with res. embedding) | 0.70 $\pm$ 0.18        | 0.69 $\pm$ 0.20        | 0.73 $\pm$ 0.19        | 1.34 $\pm$ 1.30        |

in a mixture.<sup>6</sup> The resulting phrases are mapped onto the taxonomy with a deterministic lexical procedure provided in our code. We also evaluate a variant of SEG that is additionally conditioned on the residual embedding generated at the previous iteration, which summarizes the stems yet to be extracted.

For each 10-s test segment, we compare the predicted labels  $\hat{S}$ , with one per claimed stem, against the active stem labels  $S$ , computing precision  $P = |S \cap \hat{S}|/|\hat{S}|$ , recall  $R = |S \cap \hat{S}|/|S|$ ,  $F_1 = 2PR/(P + R)$ , and the count error  $\Delta K = ||\hat{S}| - |S||$ , which reflects the difference in number of predicted stems, independent of their labels. All metrics are averaged over segments in Table 1.

As shown in Table 1, SEG achieves the highest precision and the lowest  $\Delta K$  at both levels, with an  $F_1$  on par with the multi-label classifier (0.88 vs. 0.89 for groups, and 0.74 vs. 0.75 for instruments). MusicFlamingo is competitive at the group level ( $F_1 = 0.82$ ), but degrades sharply at the instrument level ( $F_1 = 0.33$ ), suggesting that fine-grained stem identification remains challenging for language models. The oracle reaches  $F_1$  of 0.96 (groups) and 0.86 (instruments), showing part of the error stems from the classifier. Conditioning on the previous residual embedding degrades all metrics for SEG, which we attribute to exposure bias: the model is trained on GT residual embeddings but receives its own estimates at inference.

### 4.2. Source separation objective evaluation

Having assessed which stems SEG discovers, we now evaluate the separation performance when using generated queries. Each separated stem is scored against the GT stem matching the label assigned by the stem classifier. To avoid comparing mislabeled outputs, we restrict the evaluation to stems detected by SEG and, at the group level, also by MusicFlamingo. At the group level, we further restrict the evaluation to vocals, drum, bass, guitar, and piano (VDBGP), to compare with static separators trained on these stems. This leaves us with 3422 group stems and 5197 individual instruments.

We compare against two static 6-stem separators: the publicly available HTDemucs [5], and a BS-LoCoformer trained on vocals, drums, bass, guitar, piano, and other (VDBGPO) with our training data. As query-based baselines, we evaluate Banquet [9] with queries taken from a different song containing the same stem, following its original evaluation protocol, and SAM-Audio [14] with text prompts given either by the taxonomy class names (oracle) or, only at the group level, by the phrases produced by MusicFlamingo in Section 4.1 (blind). Our BS-LoCoformer and DiT-Sep separators are evaluated

<sup>6</sup>Group level: “Which instruments and voices can you hear in this recording? Answer with a comma-separated list.” Instrument level: “List every instrument and voice you can hear as specifically as possible. Break a drum kit into its separable pieces. Answer with a comma-separated list.”

**Table 2.** Objective metrics for source separation performance. Scored only on sources detected by all blind methods (mean  $\pm$  std.). Best result per block in bold.

|                  | Setting         | Separator             | Judge $\uparrow$      | SNR (dB) $\uparrow$  | MERT-MSE $\downarrow$  |
|------------------|-----------------|-----------------------|-----------------------|----------------------|------------------------|
| Groups:<br>VDBGP | Reference       | Mixture               | 1.2 $\pm$ 0.41        | -4.9 $\pm$ 8.3       | 0.22 $\pm$ 0.05        |
|                  |                 | Ground truth          | 4.7 $\pm$ 0.47        | $\infty$             | 0                      |
|                  |                 | Ground truth (AE)     | 4.7 $\pm$ 0.47        | 14 $\pm$ 9.6         | 0.04 $\pm$ 0.02        |
|                  |                 | Ideal Ratio Mask      | 4.4 $\pm$ 0.58        | 10 $\pm$ 6.6         | 0.07 $\pm$ 0.03        |
|                  | Static          | HTDemucs              | 4.4 $\pm$ 0.67        | <b>9.7</b> $\pm$ 5.0 | 0.12 $\pm$ 0.05        |
|                  |                 | BS-LoCoformer         | 4.1 $\pm$ 0.80        | 8.0 $\pm$ 4.0        | 0.14 $\pm$ 0.05        |
|                  | Oracle text     | SAM-A.                | 4.4 $\pm$ 1.01        | 4.0 $\pm$ 6.7        | 0.15 $\pm$ 0.06        |
|                  | Oracle emb.     | BS-LoCoformer         | 4.3 $\pm$ 0.69        | 8.4 $\pm$ 4.8        | 0.13 $\pm$ 0.05        |
|                  |                 | DiT-Sep               | <b>4.7</b> $\pm$ 0.49 | 6.6 $\pm$ 5.3        | <b>0.11</b> $\pm$ 0.05 |
|                  | Other song emb. | Banquet               | 4.1 $\pm$ 1.04        | 7.6 $\pm$ 7.1        | 0.14 $\pm$ 0.06        |
|                  | Blind           | MusicFlam. + SAM-A.   | 4.2 $\pm$ 1.24        | 2.7 $\pm$ 7.1        | 0.16 $\pm$ 0.07        |
|                  |                 | MuS3D (BS-LoCoformer) | 4.3 $\pm$ 0.70        | 8.1 $\pm$ 4.7        | 0.13 $\pm$ 0.05        |
|                  |                 | MuS3D (DiT-Sep)       | <b>4.7</b> $\pm$ 0.45 | 6.5 $\pm$ 5.1        | 0.12 $\pm$ 0.05        |
| All Instr.       | Reference       | Mixture               | 1.3 $\pm$ 0.41        | -9.1 $\pm$ 8.3       | 0.25 $\pm$ 0.05        |
|                  |                 | Ground truth          | 4.5 $\pm$ 0.59        | $\infty$             | 0                      |
|                  |                 | Ground truth (AE)     | 4.6 $\pm$ 0.54        | 13 $\pm$ 10          | 0.04 $\pm$ 0.02        |
|                  |                 | Ideal Ratio Mask      | 4.2 $\pm$ 0.76        | 9.1 $\pm$ 5.4        | 0.10 $\pm$ 0.05        |
|                  | Oracle text     | SAM-A.                | 4.3 $\pm$ 1.18        | -2.4 $\pm$ 8.9       | 0.21 $\pm$ 0.09        |
|                  | Oracle emb.     | BS-LoCoformer         | 4.1 $\pm$ 0.84        | <b>7.5</b> $\pm$ 4.9 | 0.15 $\pm$ 0.06        |
|                  |                 | DiT-Sep               | <b>4.6</b> $\pm$ 0.56 | 6.1 $\pm$ 6.4        | <b>0.12</b> $\pm$ 0.06 |
|                  | Blind           | MuS3D (BS-LoCoformer) | 4.1 $\pm$ 0.83        | 6.8 $\pm$ 5.5        | 0.16 $\pm$ 0.06        |
|                  |                 | MuS3D (DiT-Sep)       | <b>4.6</b> $\pm$ 0.56 | 5.6 $\pm$ 6.6        | 0.14 $\pm$ 0.07        |

both with oracle queries extracted from the GT stems and with the queries generated by SEG, the latter, corresponding to the proposed MuS3D. As references, we report the mixture, the ideal ratio mask [29], and the GT stem after encoding and decoding with the DAC-VAE autoencoder used in the DiT-Sep, Ground truth (AE).

We measure separation quality with three metrics, all computed on mono signals. The SAM-Audio Judge [30] is a learned, reference-free model that predicts perceptual quality scores for a separated stem given the mixture and a text prompt as context, for which we use the taxonomy label of the target; we report its ‘‘Overall’’ score. We report the signal-to-noise ratio (SNR), which compares waveforms and therefore penalizes imperceptible phase differences, such as those introduced by the autoencoder of the DiT-Sep. We also report the mean-squared error between MERT representations [31] (12th layer, following [32]), a perceptually motivated metric that correlates with subjective DMOS ratings and is insensitive to slight phase differences; we refer to it as MERT-MSE.

As shown in Table 2, both separators (BS-LoCoformer, DiT-Sep) perform almost equally well with SEG queries and oracle queries at both levels: the Judge score is unchanged, and MERT-MSE increases by at most 0.01, so the separators are robust to estimation errors from SEG. This also holds at the instrument level, where SEG generates more numerous, finer-grained queries, and MuS3D matches the group-level scores. Remarkably, MuS3D (DiT-Sep) obtains the highest Judge score at both levels (4.7 and 4.6), on par with GT stems.

SAM-Audio scores lower, even with oracle text prompts, and degrades further with MusicFlamingo prompts, often extracting the wrong stem. HTDemucs obtains the highest SNR and a MERT-MSE comparable to the best query-based separator, but a lower Judge score than the DiT-Sep. The DiT-Sep obtains a comparatively low SNR as it generates waveforms from latents; the higher SNR of autoencoded GT stems (14 and 13 dB) shows the autoencoder is partly responsible.

### 4.3. Source separation subjective evaluation

For the subjective evaluation, we follow the pairwise comparison experiment setup used in [33]. Trials use 4-s excerpts from the MoisesDB test set. We restrict the taxonomy to VDBGP, since this performed the most consistently across the models. Each trial presents

|                  |     |    |                     |             |
|------------------|-----|----|---------------------|-------------|
| MuS3D (DiT-Sep)  | 116 | 14 | HTDemucs            | $p < 0.001$ |
| MuS3D (DiT-Sep)  | 109 | 21 | MusicFlam. + SAM-A. | $p < 0.001$ |
| MuS3D (DiT-Sep)  | 59  | 71 | Oracle (DiT-Sep)    | $p = 0.34$  |
| MuS3D (DiT-Sep)  | 61  | 69 | Ground truth (AE)   | $p = 0.54$  |
| MuS3D (DiT-Sep)  | 44  | 86 | <b>Ground truth</b> | $p < 0.001$ |
| Oracle (DiT-Sep) | 68  | 62 | Ground truth (AE)   | $p = 0.66$  |
| Oracle (DiT-Sep) | 41  | 89 | <b>Ground truth</b> | $p < 0.001$ |

**Fig. 2.** Pairwise preference listening test results from 130 trials per pair. Green/red indicate a significant preference ( $p < 0.05$ ), and the winning method is highlighted in bold in such cases.

the participant with the song mixture and two examples of separated stems. The task is to pick the option that better separates the indicated source, i.e., vocals, bass, drums, guitar, or piano. The listening test involved 13 participants with no reported hearing problems and with prior listening test experience, conducted in soundproof booths of the Aalto Acoustics Lab using headphones at constant loudness. The test used 2 songs per instrument, shared across all compared pairs to avoid selection bias. SAM-Audio was prompted with the phrases produced by MusicFlamingo, and all excerpts were taken from stems detected by both MusicFlamingo and SEG, so that separation errors are attributable to the separator.

Fig. 2 shows the results for the subjective listening test. For each pair, we test whether the number of times a condition was chosen differs from random chance with a two-sided binomial test, treating  $p < 0.05$  as significant. The comparisons show that MuS3D (DiT-Sep) significantly outperforms the baselines (HTDemucs and SAM-Audio), achieves parity with Oracle (DiT-Sep) and Ground truth (AE), but is significantly outperformed by the original, unencoded stem. Oracle (DiT-Sep) and Ground truth (AE) are not significantly different, but both are outperformed by the Ground truth. This indicates that the limiting factor is the autoencoder, not the separation or query generation method.

## 5. CONCLUSION

This paper proposes MuS3D, a framework for blind music source separation that automatically discovers the stems in a mixture using audio-query generation. Its SEG module recursively generates one embedding per stem, conditioned on the mixture and on the previously generated embeddings of the other stems. The embeddings can then condition a query-based separator, so no text or audio example has to be supplied by the user.

The SEG outperforms a language-model baseline at identifying the sources in a mixture. On correctly detected sources, separators conditioned on the generated queries perform on par with those conditioned on GT queries, in both objective metrics and a formal listening test. The listening test results also suggest that, for the DiT-Sep separator, the audio autoencoder is the primary limiting factor on perceived quality. This paper shows that audio queries can be generated automatically without loss in separation quality, enabling MSS when the stems in a mixture are not known in advance.

That said, we identify limitations that should be addressed in future work. SEG still misclassifies or misses some stems, which lowers recall (Table 1). The evaluation is limited to 10-s segments; extending it to whole songs, where queries must stay consistent across segments, remains open. Finally, the inference pipeline, which samples from SEG and then runs DiT-Sep, is computationally expensive, but techniques such as flow maps [34] offer a route for acceleration.

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